A COUNTEREXAMPLE TO THE ETZION–VARDY–YAAKOBI CONFERENCE-MATRIX CONJECTURE

Abstract. Etzion, Vardy, and Yaakobi conjectured that, for prime p, every conference matrix of order $p+1$, reduced modulo p, generates an MDS code over $\mathbb{F}_p$. We refute their Conjecture 5 at $p = 29$. An explicit symmetric conference matrix of order 30 generates a Euclidean self-dual $[30, 15]_{29}$ code containing a word of Hamming weight 13; hence its minimum distance is at most 13, rather than 16.

1. The counterexample

A conference matrix of order n is a matrix with zero diagonal, off-diagonal entries in $\{\pm 1\}$, and

$$WW^{\mathsf{T}} = (n-1)I_n.$$

Etzion, Vardy, and Yaakobi conjectured that, for prime p, the code over $\mathbb{F}_p$ generated by any conference matrix of order $p+1$ is MDS with parameters

$$\left[p+1, \frac{p+1}{2}, \frac{p+3}{2}\right]_p.$$

This is Conjecture 5 of [1]; the statement carries the same number in the openly available preprint version [2], whose concluding section also lists it among the questions left open.

Status. Etzion, Vardy, and Yaakobi report that the conjecture was verified up to $n = 23$, for conference matrices arising from Paley's quadratic-residue construction [1, 2]. The order 30 treated here lies outside that range, so what follows contradicts the conjecture and not the reported verification: the conjecture is asserted for *every* conference matrix of order $p+1$, and one such matrix whose row span is not MDS therefore settles it as stated. Nothing here decides the status of the other conference matrices of order 30, or of Paley's construction at larger orders. Nor are the conjectured parameters an empty target: [1] notes that self-dual MDS codes of length $q+1$ over $\mathbb{F}_q$, q a prime power, were constructed by Grassl and Gulliver [3]. What fails at $p = 29$ is the assertion that a conference matrix of order $p+1$ always generates one.

In the following display, $0, +, -$ denote $0, 1, -1$, respectively. Let W be the symmetric matrix whose rows are

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+0--------------++++++++++++++
+-0------+++++++-------+++++++
+--0----+-++++++-++++++------+
+---0+++----+++++---+++---+++-
+---+0++-+-+--++--++-+++++----
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Theorem 1. *The code*

$$C = \mathrm{rowspan}_{\mathbb{F}_{29}}(W)$$

is a Euclidean self-dual $[30, 15]_{29}$ *code satisfying* $d(C) \leq 13$. *Consequently,* C *is not MDS, and Conjecture 5 of Etzion–Vardy–Yaakobi is false.*

Proof. Exact integer multiplication gives

$$WW^{\mathsf{T}} = 29 I_{30},$$

so W is a conference matrix. After reduction modulo 29, its rows are mutually orthogonal. Thus $C \subseteq C^{\perp}$, and therefore $\dim C \leq 15$. On the other hand, the determinant of the leading 15×15 principal submatrix of W is

$$-8584878 \equiv 21 \pmod{29}.$$

Hence $\mathrm{rank}_{\mathbb{F}_{29}}(W) \geq 15$. It follows that $\dim C = 15$ and $C = C^{\perp}$.

Let

$$x = (25, 5, 26, 12, 0, 6, 19, 16, 0, 22, 5, 1, 26, 23, 14, \underbrace{0, \ldots, 0}_{15}) \in \mathbb{F}_{29}^{30}.$$

Direct multiplication in $\mathbb{F}_{29}$ gives

$$xW = (1, \underbrace{0, \ldots, 0}_{14}, 15, 6, 0, 8, 27, 0, 18, 14, 0, 15, 4, 11, 20, 17, 19).$$

Its support is

$$\{1, 16, 17, 19, 20, 22, 23, 25, 26, 27, 28, 29, 30\},$$

so it has Hamming weight 13. Hence $d(C) \leq 13$. By the Singleton bound, an MDS $[30, 15]_{29}$ code has minimum distance $30 - 15 + 1 = 16$. Therefore C is not MDS. $\square$

Provenance. The displayed matrix is reconstructed from representative 3 in Spence's order-30 conference two-graph data file [4]. This provenance is not used in the proof; the matrix, minor, and codeword above form a complete finite certificate.

Independent re-run. The computation was re-run independently, on a separate machine and by a standalone program written against this manuscript, and it agreed: all 43 checks passed. The program and its transcript accompany this note. Its mathematical inputs are transcribed from this text and are nothing else: the 30 printed rows of W, the vector x, the printed product xW with its support, the minor -8584878 and its residue 21, the weight 13, the Singleton value 16, and the parameters $[30, 15]_{29}$. Everything else it recomputes in exact integer and $\mathbb{F}_{29}$ arithmetic, with no floating point anywhere. The checks cover the decoding of the display and its zero diagonal, ±1 off-diagonal entries and symmetry; all 900 integer inner products of rows, that is $WW^{\mathsf{T}} = 29I_{30}$, so that W is verified to *be* a conference matrix of order 30 rather than assumed to be one, together with the independent consequence $|\det W| = 29^{15}$; the leading 15×15 minor -8584878 by fraction-free Bareiss elimination and again by elimination over the rationals, and its residue 21; $\mathrm{rank}_{\mathbb{F}_{29}}(W) = 15$; the equality $C = C^{\perp}$, obtained by computing the $\mathbb{F}_{29}$ null space of W and checking both containments, the second of them also by solving $y = zW$ against the raw rows of W; the product xW recomputed entry by entry, with its support and weight 13; the same weight-13 word re-derived from the reduced row echelon form of W without using x at all; and the Singleton value $30 - 15 + 1 = 16$ together with the strict inequality $13 < 16$. Every number in the theorem and its proof was reproduced.

Five limits of that re-run should be recorded, each of them noted in the program's own output. First, the exact minimum distance of C was not determined. An exhaustive census of the codewords whose information vector in the systematic coordinates has weight at most 5, namely 52 531 811 444 of the $29^{15} - 1$ nonzero words, returns minimum weight 12; words of information weight 6 or more were not enumerated. The bound $d(C) \leq 13$ proved above is therefore correct but not tight, which does not affect the refutation, since $13 < 16$ already. Second, no census over the conference matrices of order 30 was attempted, and none is required: the theorem concerns the matrix exhibited above and asserts nothing about the others. Third, the provenance sentence was not checked, the program being self-contained and fetching nothing. Fourth, four of the 43 checks test the program's own machinery rather than the paper, and cannot fail on a well-formed input. Fifth, the checks are invariant under the sign switchings $W \mapsto DWD$ with $D = \mathrm{diag}(\pm1)$ whose -1 entries are confined to the coordinates where x and xW both vanish, here coordinates 5, 9, 18, 21 and 24: an orbit of $2^5 = 32$ displays. Every member of that orbit is again a conference matrix whose $\mathbb{F}_{29}$ row span is a self-dual $[30, 15]_{29}$ code containing the same weight-13 word, so the theorem holds verbatim for each of them, and within the orbit it is only the provenance sentence that singles out the display printed above; the program prints its decoded matrix back so that the display above can be compared with it character by character. An independent re-run that agrees is evidence about the computation, not a second proof.

References

[1] T. Etzion, A. Vardy, and E. Yaakobi, *Coding for the Lee and Manhattan metrics with weighing matrices*, IEEE Trans. Inform. Theory **59** (2013), no. 10, 6712–6723, doi:10.1109/TIT.2013.2268156.

[2] T. Etzion, A. Vardy, and E. Yaakobi, *Coding for the Lee and Manhattan metrics with weighing matrices*, preprint, arXiv:1210.5725v2 (2012), `https://arxiv.org/abs/1210.5725v2`. Conjecture 5 of this version is the statement refuted here.

[3] M. Grassl and T. A. Gulliver, *On self-dual MDS codes*, Proc. IEEE International Symposium on Information Theory (ISIT), Toronto, 2008, 1954–1957, doi:10.1109/ISIT.2008.4595330.

[4] E. Spence, *Conference two-graphs*, data collection, University of Glasgow, index page `https://www.maths.gla.ac.uk/~es/twograph/conf2Graph.php`; the order-30 data are the file `conf2graph.30` listed there, `https://www.maths.gla.ac.uk/~es/twograph/conf2graph.30`, accessed August 23, 2026.